

SPICKARD, MO, USA

WMO: 725400

Lat: 40.250N

Lon: 93.720W

Elev: 889

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 82-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.1	(c) 4.8	(d) -11.7	(e) 3.0	(f) 2.9	(g) -6.0	(h) 4.1	(i) 6.4	(j) 26.9	(k) 28.6	(l) 24.5	(m) 29.2	(n) 7.7	(o) 0	(p) 0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	96.0	74.5	92.4	73.7	89.1	73.2	78.3	88.4	76.6	87.2	75.1	85.6	10.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	138.6	84.2	73.5	128.7	82.5	71.9	121.7	80.7	42.4	88.0	40.6	87.5	39.1	85.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
24.0	20.4	18.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-4.1	79.7	8.6	1.9	-10.3	81.1	-15.3	82.2	-20.1	83.3	-26.4	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	27.9	32.1	42.9	51.9	63.4	72.4	77.1	74.8	65.4	54.2	40.7	31.1
	DBStd	19.44	11.17	12.28	11.12	9.32	7.00	6.64	5.59	7.00	8.90	8.76	10.26	11.02
	HDD50	2508	687	505	274	84	2	0	0	0	4	55	308	589
	HDD65	5534	1151	921	689	406	113	15	1	8	102	348	729	1051
	CDD50	3587	1	4	54	141	418	671	841	770	466	186	31	4
	CDD65	1135	0	0	3	13	64	236	376	313	115	14	1	0
	CDH74	10928	0	0	38	148	507	2209	3895	2995	988	142	6	0
	CDH80	4477	0	0	10	35	105	862	1747	1341	350	27	0	0
Wind	WSAvg	8.8	9.4	8.8	10.5	10.4	8.9	8.0	7.6	7.0	7.9	8.5	9.6	8.9
Precipitation	PrecAvg	35.7	1.1	1.0	2.4	3.4	4.9	4.3	4.8	4.0	4.0	3.0	2.0	1.5
	PrecMax	55.2	3.9	3.5	7.5	6.9	15.5	10.9	21.3	10.5	10.9	8.3	6.4	5.5
	PrecMin	15.9	0.0	0.1	0.0	0.6	1.1	0.4	0.7	0.0	0.7	0.2	0.0	0.0
	PrecStd	8.5	0.9	0.8	1.7	1.7	3.0	2.5	3.9	2.2	2.4	1.9	1.5	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.3	65.1	81.1	84.9	87.0	96.9	100.2	100.7	95.1	83.2	73.6	62.5
		MCWB	50.1	51.5	61.0	66.7	69.8	73.0	75.8	75.6	74.6	65.7	60.0	55.5
	2%	DB	51.5	57.6	71.3	78.5	82.7	91.6	95.6	96.0	89.1	77.9	66.0	55.6
		MCWB	43.1	48.4	56.7	62.5	67.6	71.8	74.8	74.5	70.4	62.8	57.3	50.5
	5%	DB	46.6	52.4	65.6	73.0	79.4	88.0	92.3	91.8	83.5	73.2	60.8	49.8
		MCWB	40.5	44.1	53.6	58.0	65.8	73.1	74.2	74.3	69.7	60.9	53.0	44.2
	10%	DB	42.0	47.9	60.4	68.0	76.3	84.7	89.1	87.3	79.5	68.8	56.4	44.3
		MCWB	36.7	41.5	51.2	55.8	63.8	71.2	73.3	73.0	67.9	58.5	49.7	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	52.9	63.0	67.3	72.7	79.1	82.0	79.1	75.7	68.0	62.2	58.6
		MCDB	56.7	63.5	78.1	81.0	83.6	86.5	89.9	92.0	88.2	79.9	70.2	61.4
	2%	WB	44.2	48.5	58.8	63.7	69.4	76.7	79.3	77.4	73.7	65.0	58.8	51.1
		MCDB	50.4	56.8	68.6	75.5	79.5	85.3	88.5	89.7	85.2	73.9	64.8	54.1
	5%	WB	40.7	44.8	55.5	60.2	67.6	74.8	77.2	75.8	71.7	62.7	54.2	43.8
		MCDB	46.5	51.7	64.3	69.4	77.4	84.1	87.0	87.2	80.9	71.0	60.1	49.5
	10%	WB	37.3	41.8	51.5	56.7	65.4	73.0	75.3	74.3	69.3	59.9	50.0	39.9
		MCDB	41.3	47.3	60.0	66.1	73.6	82.5	86.2	84.8	77.7	67.7	55.8	44.0
Mean Daily Temperature Range	5% DB	MDBR	16.2	17.0	18.9	20.0	19.0	19.6	20.2	19.9	20.7	20.5	17.3	15.8
		MCDBR	21.8	23.9	26.9	27.9	22.8	22.4	23.0	24.8	23.5	25.7	22.7	23.0
		MCWBR	15.9	16.4	16.7	16.6	13.1	10.9	9.9	9.4	11.4	15.0	16.0	17.2
	5% WB	MCDBR	21.0	22.4	25.3	23.7	19.6	19.6	19.7	21.0	19.8	22.1	19.4	21.3
MCWBR		15.5	15.8	17.1	15.5	12.7	10.5	10.0	9.5	10.7	14.2	15.1	16.8	
Clear-Sky Solar Irradiance	taub	0.292	0.318	0.338	0.389	0.419	0.421	0.435	0.431	0.381	0.340	0.298	0.289	
	taud	2.441	2.365	2.392	2.266	2.249	2.281	2.240	2.271	2.359	2.470	2.527	2.475	
	Ebn at Noon	278	284	291	280	273	272	267	265	272	273	275	270	
	Edh at Noon	25	31	34	42	43	42	43	41	35	28	23	23	
All-Sky Solar Radiation	RadAvg	636	921	1198	1535	1803	2005	2014	1757	1503	1026	706	551	
	RadStd	64	76	99	103	150	178	143	95	96	115	62	58	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page